

formance and perform multiple test procedures at the physical and various software layers. These test systems can also be used to understand the complex interactions between SDR subsystems in the frequency and time domains, especially in burst or frequency-hopping signals.

When filtered and amplified, software anomalies can create temporal RF impulse bursts of energy at the RF output. To isolate software and hardware performance, RTSAs can be used to trigger on transients in the frequency domain, capture the events into memory and drive the other test instruments to probe possible error sources. The acquired signals are presented in a time-correlated fashion, helping designers see how anomalies in the digital and analogue blocks of an SDR will propagate to the RF output as impulse noise.

The unique ability of these RT-SAs to find problems from spectral transients can be used to trigger the other instruments and obtain time-correlated views of vastly different hardware and software functional implementations. For example, the RTSA can capture the signal in the RF and IF portions of the signal paths, and a logic analyser can capture the digital baseband signal and compare it to the symbol table produced by the RTSA. Furthermore, some RTSAs offer offline software that can be used to analyse acquired data from the logic analyser and oscilloscope, allowing hardware and software measurement correlation.

Verifying baseband IQ waveform quality

Verifying baseband IQ waveform quality is important to both system engineers and field-programmable gate array (FPGA) designers. It helps engineers test the baseband to ensure it is properly functional at an early stage of development because many of the problems involved in digital circuits are in the FPGA design.

The baseband signals in the

actual designs and applications are differential (I+, I-, Q+ and Q-) and may possess a DC offset. In the past, very few spectrum analysers were able to test IQ signals directly, and fewer spectrum analysers could test the baseband IQ signals with DC offset. Engineers have been forced to use oscilloscopes with additional software for post analysis.

Select RTSAs enable baseband IQ testing using differential inputs. Doing so delivers measurement consistency when analysing IQ, IF and RF signals. Testing IQ signals with an RTSA also reduces system complexity and simplifies testing procedures, while offering higher dynamic range and greater memory depth than general-purpose instruments.

Modern RTSAs bring baseband, RF and post-analysis functionality together. For example, pre-eminent RTSAs can perform DC baseband measurements with 14-bit analogue-to-digital conversion (ADC), ensuring measurement accuracy. Some also possess the differential IQ input function, which enables engineers to connect the RTSA directly to baseband IQ signals for error vector magnitude (EVM) analysis—without any additional differential probe set. In addition to EVM, these RTSAs provide fully time-correlated measurements across multiple domains (time domain, frequency domain, modulation domain and constellation). This capability can be invaluable for troubleshooting frequency-hopping SDRs.

Frequency settling time measurements for hopped signals

Frequency settling time defines the length of time between two hopped

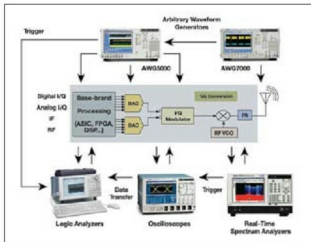


Fig. 3. An integrated, end-to-end test system for verifying and troubleshooting SDRs, featuring the real-time spectrum analyser (RTSA), arbitrary waveform generator (AWG), oscilloscope and logic analyser

frequencies. It is one of the primary contributors to a frequency-hopping system's efficiency. The shorter the frequency settling time, the faster a system can hop. Measuring the frequency settling time ensures optimal synthesiser operation and maximises overall system performance.

The traditional way of measuring frequency settling time was limited by the instrumentation and was very time consuming. Engineers were forced to rely on oscilloscopes and frequency discriminators for the test, showing only the signal envelope and hinting at the stability of the signals. While oscilloscopes have excellent timing resolution, using them to measure small frequency changes can be challenging (depending on the frequency resolution required for the measurement). Oscilloscopes cannot automatically measure hopped frequencies, and frequency settling time can only be estimated.

Leading RTSAs offer automated frequency settling time measurements. By setting parameters such as frequency settling threshold and smoothing factor, engineers can measure the frequency settling time for hopped signals quickly and accurately. Engineers can also see the spectrum changes during the hops.

In addition to time-correlated measurements across multiple do-